SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club | Submission deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model and sponsor resource used to build the project. Complete every section; where a category does not apply, write **No** or **None** - do not leave it blank. Submit as a single PDF named **SDC2026_TeamName_TechDeclaration.pdf**.

1. Team & Project Overview

Field	Response
Team name	Empathy Coders
Project / solution name	OrderPilot - Smart Order Consolidation and Approval for SMEs
Name (leader)	Yang Jiahui
Email & phone (leader)	[TO BE COMPLETED BEFORE SUBMISSION]
University affiliation(s)	Xiamen University Malaysia (XMUM)
Live prototype URL (Replit)	https://order-pilot--microtopt.replit.app
Code repository URL	https://github.com/Jacky-Godlike/Empathy-Coders
Replit project	[COPY FROM THE REPLIT ADDRESS BAR]
RunPod	Not applicable - no GPU training, fine-tuning or inference; the RunPod credit was not used

2. Architecture & Technology Stack

2.1 System Architecture Overview [maximum 200 words]

OrderPilot is a Node.js/Express web application served by Replit. The browser UI (HTML/CSS/vanilla JavaScript) calls a REST API. The API orchestrates four services: triage, order/approval, inventory and activity. Triage sends synthetic message text to a server-side AI adapter, which calls the configured provider (DeepSeek in the submitted demo) and returns a constrained JSON extraction. If the AI call fails or no key is configured, a deterministic rules extractor produces the same shape. The order service consolidates follow-up messages, matches the customer and catalogue, and calls a pure decision engine. The decision engine reads products, stock, suppliers and existing orders, then calculates stock availability, shortage, supplier lead-time feasibility, revenue, cost, gross profit, margin, price deviation, duplicates and missing fields. A human owner/manager approves; approval reserves stock and creates internal purchase drafts through a mock OrbyteAdapter. Every action writes an activity entry. Data is stored in JSON files (seed committed, runtime gitignored) behind a repository layer, so a SQLite/Postgres or real Orbyte implementation can replace it without changing the services.

2.2 Languages, Frameworks & Libraries

Layer	Technology	Purpose
Frontend	HTML5, CSS3, vanilla JavaScript (ES modules)	Order Desk, Inbox, Orders, Activity, Catalog, Settings
Backend	Node.js 20+, Express 4	REST API, RBAC, triage orchestration, exports
Data	JSON files (seed committed; runtime gitignored)	Customers, products, stock, suppliers, messages, orders, drafts, activity
AI	DeepSeek chat/completions API (OpenAI-compatible)	Classification and field extraction only
Fallback	Deterministic rules extractor	Keeps the demo working without an API key
Exports	ExcelJS + CSV writer	Order and activity exports
Hosting	Replit (Core credit)	Required deployment platform
Repository	GitHub, Apache License 2.0	Public source and commit history

3. APIs & Cloud Services

3.1 External APIs

API and provider	Purpose in project	Access tier
DeepSeek API - chat/completions (OpenAI-compatible)	Classify synthetic messages and extract order fields only. Not used for stock, margin or risk.	Team-paid / free tier
Google Fonts (fonts.googleapis.com, fonts.gstatic.com)	Web font delivery (DM Sans, Fraunces). The UI falls back to system fonts if unavailable.	Free

No maps, payments, messaging, banking, government or real Orbyte APIs are called. The WhatsApp/email orders are synthetic messages pasted or uploaded into the app; no real WhatsApp/email API is connected. No API keys appear in this document.

3.2 Cloud Platforms & Deployment

Service and provider	Purpose in project	Plan / region
Replit	Development, runtime, Secrets, deployment and Webview	Replit Core (sponsor-provided credit)
GitHub	Public source repository, commit history, Apache 2.0 license	Free
DeepSeek	Server-side inference API	Team-paid / free tier

The app is deployed with Replit Deployments from the GitHub repository. The .replit file runs npm start (Node 20). Static assets and the API are served by Express; there is no separate build step. Runtime JSON data is created on boot from the committed seed. No cloud database, object storage, authentication service or payment service is used.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

Model & version	Source / provider	Usage	License / terms
deepseek-flash (DeepSeek-V4.1-Flash)	DeepSeek	Pre-trained; called via API for message classification and field extraction. Not fine-tuned or self-trained by the team.	Provider API terms
Deterministic rules extractor	Team-created code	Fallback classification/extraction when the AI is unavailable or returns invalid output	Team code, Apache 2.0

The AI model is not used to decide stock, margin, delivery feasibility, duplicates or approval. It returns structured fields; the server validates them against the catalogue and the decision engine recalculates every business rule. The model does not receive real personal data; all demo messages are synthetic.

4.2 Datasets

Dataset name	Type	Source / URL	Personal data?	License
OrderPilot synthetic SME seed	Synthetic	Team-created JSON in data/seed/	No - fictional names, phones, emails and addresses	Team-created; Apache 2.0
OrderPilot synthetic message test set	Synthetic	Team-created; 29 WhatsApp/email-style messages	No	Team-created; Apache 2.0
Google Fonts	Font assets (not training data)	fonts.google.com	No	SIL OFL 1.1

No public or private dataset was used for training, fine-tuning or evaluation of the AI model. No real customer, supplier, company or Orbyte data is included. The synthetic messages cover clear orders, Malay/Manglish, add-ons, corrections, ads, inquiries, supplier pitches, payment reminders and spam.

5. AI-Assisted Development Tools

Did your team use any generative-AI tool to help create code, design, text, or other project material?

[x] Yes

[ ] No

Tool & version	Used for	Extent of use
Claude Code (Anthropic)	Code scaffolding, implementation review, debugging, documentation drafts and UI design support	Moderate / major - the team directed the architecture and reviewed every file
No other generative-AI development tools declared	None	None

The team must remove tools they did not use and add any they did. Undeclared use can affect eligibility.

How the team reviewed, tested and took responsibility for AI-generated output [maximum 100 words]

The team defined the problem, the five-step journey, the recommendation rules and the approval model before using AI assistance. Every generated file was read, adapted and tested. We ran the 42-test suite and manually exercised the desktop and mobile UI, including the scan, consolidation, approval, request-info, staff-403, export and reset flows. We can explain the architecture, the decision rules, the AI/fallback split and the data model. AI was a productivity assistant; the problem framing, solution design and technical decisions are the team's.

6. Sponsor-Provided Resources

Did your project use any official SDC sponsor-provided resource?

[x] Yes - Replit Core

[ ] No

Sponsor	Resource	How it was used
Replit	Replit Core credit	Development, runtime, Secrets, deployment and live demo URL
RunPod	US$25 credit	Not used. The project has no GPU workload: no training, fine-tuning or GPU inference. The AI model is called as a hosted API.
Orbyte (Alpha Red)	Platform access / dataset	Not used. No production Orbyte access, API or customer data was provided or used. A mock OrbyteAdapter models the future integration over synthetic data.

Was the sponsor resource essential to the core functionality, or could it be replaced? [maximum 100 words]

Replit is required by the competition and is used for the live prototype and deployment; the application can also run locally with npm start, so the business logic is not locked to Replit. RunPod is not used because there is no GPU workload. Orbyte production access is not used; the OrbyteAdapter interface is the integration seam, and the mock implementation is replaceable by a real Orbyte Connect adapter without changing the order desk.

7. Security, Privacy & Responsible Use

Data Handling & Privacy [maximum 150 words]

The demo stores only synthetic customer, product, supplier, stock, message, order and activity records created by the team. No real personal, customer or company data is collected. The AI provider receives the synthetic message text and the synthetic catalogue context needed for extraction; the API key is held server-side in Replit Secrets (or a gitignored .env locally) and is never sent to the browser, logged or committed. The demo runtime data is stored in JSON files and lasts until the manager resets the demo; the reset is logged. A real deployment would apply PDPA principles: purpose limitation, data minimisation, a retention schedule, role-based access, encryption in transit, a data subject request process and a documented cross-border transfer basis. No real payments, statutory filings or supplier

purchase orders are processed.

Known Limitations & Risks [maximum 150 words]

AI hallucination or omission. The model may miss or invent a field. Mitigation: a strict JSON schema, catalogue validation, confidence scoring, mandatory missing-field checks, deterministic re-calculation of every business number, a visible source message, and human approval. Provider dependency and rate limits. Mitigation: timeout, one retry, per-message fallback to the deterministic extractor, and a UI/label that states which mode ran. Single-process JSON storage. Mitigation: a repository interface; a SQLite/Postgres store is the production path. Simulated authentication. The actor header is a demo device; the server still enforces roles. A real deployment would use Orbyte SSO and signed sessions. Replit Autoscale persistence. The demo reseeds on boot; a Reserved VM or external database is the production path. Scope. The app does not process payments, send real purchase orders, or give legal/financial advice.

8. Compliance & Readiness

[x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

[x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

[x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

[x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.

[x] Any use of official SDC sponsor resources has been declared in Section 6.

[ ] The prototype described here is accessible to judges through the links provided in Section 1.

[x] All fields in this template have been completed within the word limits stated.

The accessibility box is ticked after the final Replit deployment; the live URL is added in Section 1.

9. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature: _____ **Date:** 21 October 2026

Name: Yang Jiahui (Team Leader) **Team:** Empathy Coders

Signature: the team leader signs here before submitting. Date: 21 October 2026.